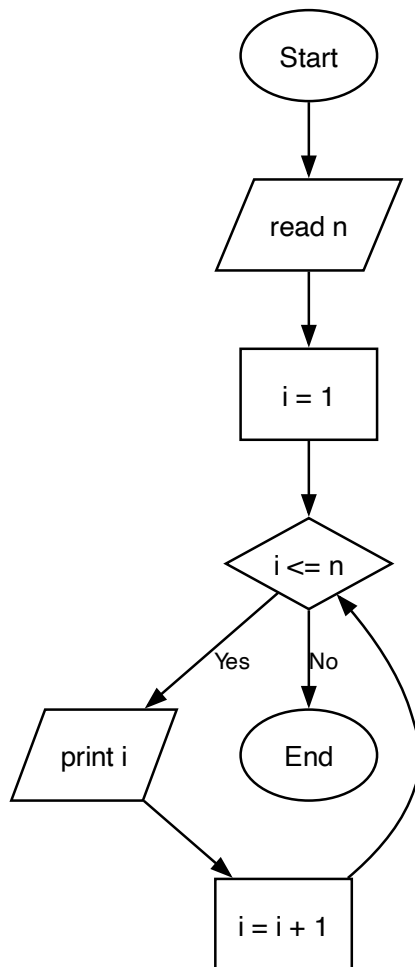


Answer key

Problem 1.

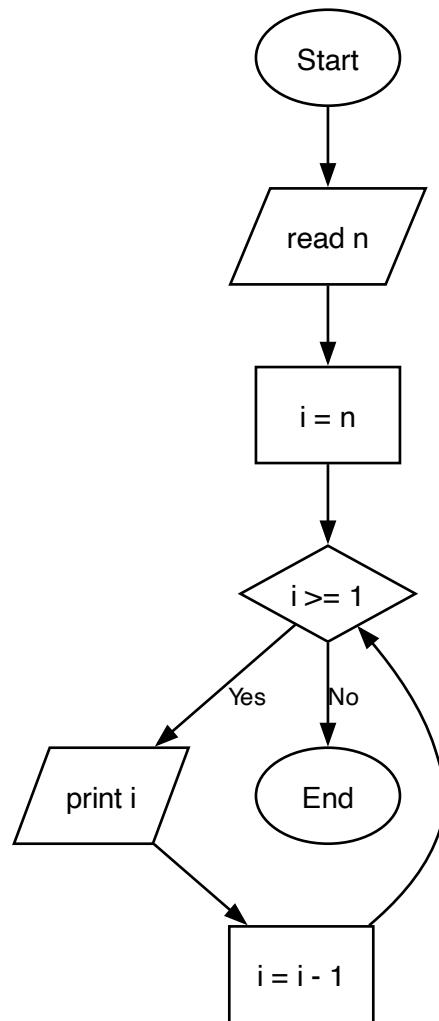
Trace this algorithm for $n = 5$, and complete the trace table.



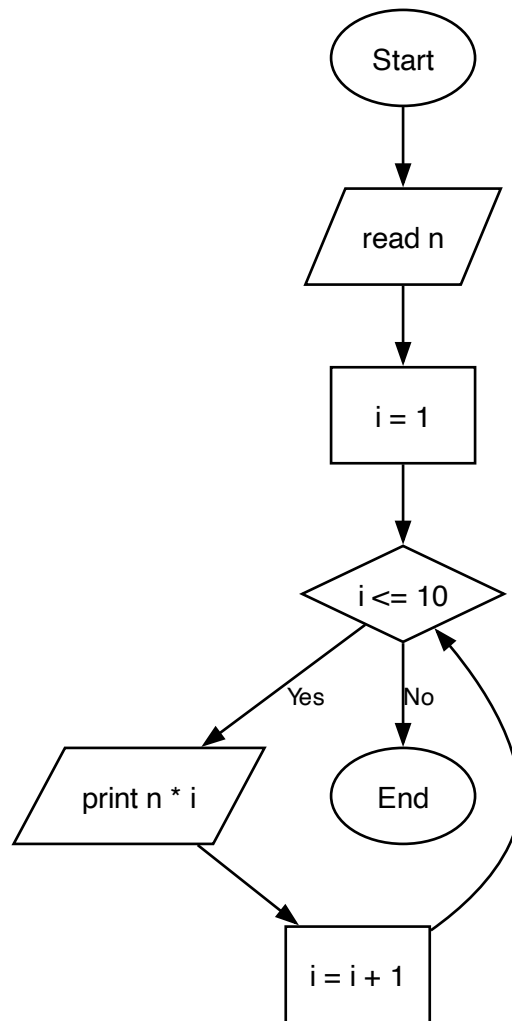
n	i	output
5		
5	1	
5	1	1
5	2	
5	2	2
5	3	
5	3	3
5	4	
5	4	4
5	5	
5	5	5
5	6	

Output: 1, 2, 3, 4, 5

Problem 2.

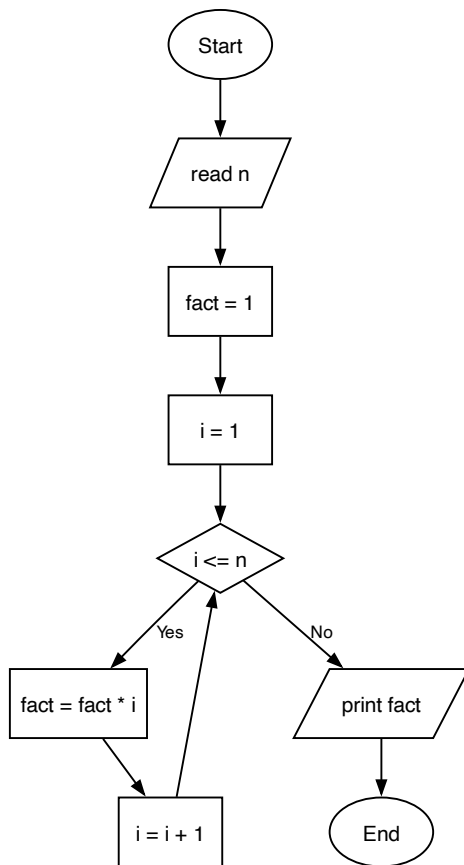


Problem 3.



Problem 4.

Trace this algorithm for $n = 5$, and complete the trace table.



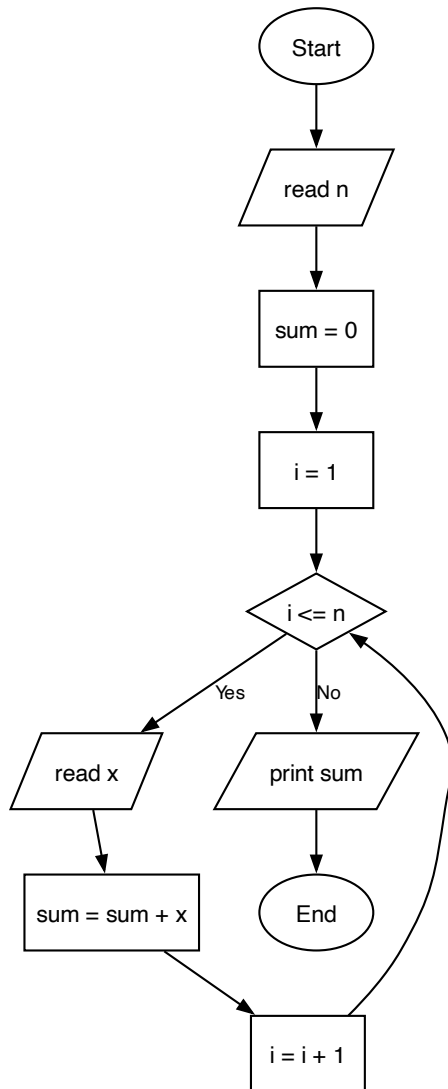
n	fact	i	output
5			
5	1		
5	1	1	
5	1	1	
5	1	2	
5	2	2	
5	2	3	
5	6	3	
5	6	4	
5	24	4	
5	24	5	
5	120	5	
5	120	6	
5	120	6	120

Output: 120

Problem 5.

This program reads how many numbers there are (n), then reads that many numbers one by one and prints their total.

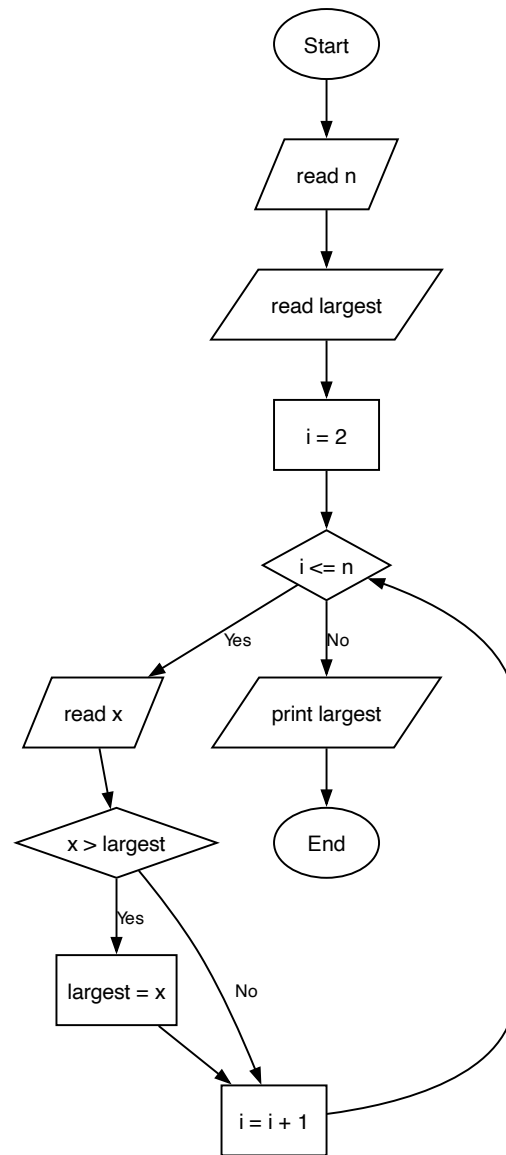
Trace this algorithm for $n = 3$, where the values of x entered are 10, 20, 30 (in that order), and complete the trace table.



n	sum	i	x	output
3				
3	0			
3	0	1		
3	0	1	10	
3	10	1	10	
3	10	2	10	
3	10	2	20	
3	30	2	20	
3	30	3	20	
3	30	3	30	
3	60	3	30	
3	60	4	30	
3	60	4	30	60

Output: 60

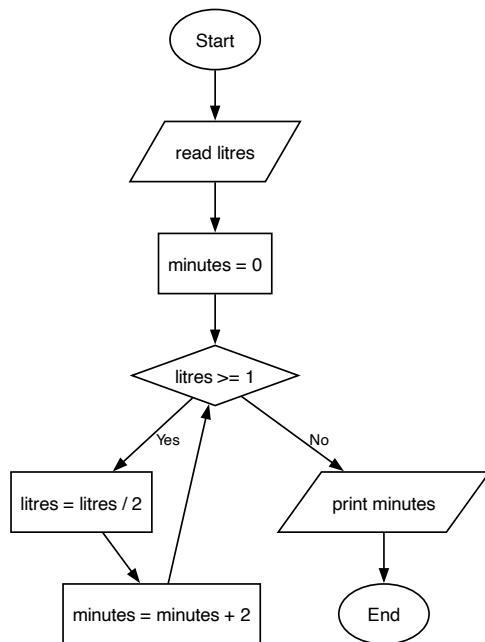
Problem 6.



Problem 7.

A water tank holds some litres of water. Every 2 minutes, exactly half of the water currently in the tank drains out. The program keeps going until less than 1 litre is left, and prints how many minutes that took. (Notice that the tank never becomes completely empty — can you see why?)

Trace this algorithm for $\text{litres} = 64$, and complete the trace table.



litres	minutes	output
64		
64	0	
32	0	
32	2	
16	2	
16	4	
8	4	
8	6	
4	6	
4	8	
2	8	
2	10	
1	10	
1	12	
0.5	12	
0.5	14	
0.5	14	14

Output: 14

Problem 8.

